

ANOOJ LAL

SOFTWARE ENGINEER

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EXPERIENCE

E*TRADE *Software Engineer*

Jan 2022 – Present (4 Years)

- Implemented several Spring Batch pipelines to process high-volumes of stock market data (~18.5M rows/day)
- Designed a fault-tolerant distributed system of batch app nodes, ensuring a “standby” node turns “active” during a crash
- Optimized the processing of options data by implementing multi-threaded concurrency, reducing processing time by 4x
- Collaborated with 3 client platform teams to deliver customer-facing data via a Spring Boot REST API
- Built an internal UI to interface our financial instrument DBs, making it easier to run complex SQL across several tables
- Added monitoring of our data vendor directories to ensure on-time arrival of files and raise alarms for missing files
- Utilized GitHub Copilot to run report comparison agents to validate features and inform optimization decisions
- Regularly led knowledge sharing sessions to demonstrate gen-AI’s role in boosting productivity by 10x

Java · XML · Spring Boot · Spring Batch · Spring Security · Copilot · Oracle DB · CyberArk · Prometheus · React · Python · Jenkins CI/CD

State Farm *Software Engineer, Enterprise Tech Intern*

Jan 2019 – Dec 2021 (2 Years)

- Maintained and enhanced Spring Boot API microservices for querying customer insurance data
- Developed extensive unit tests for Java code and API endpoints and configured CI/CD pipelines
- Wrote scripts to redact sensitive data from call transcriptions for use in sentiment analysis
- Recreated the popular ad “Attacking Buffaloes” in VR to showcase a new medium for training insurance agents

Java · Spring Boot · JUnit · Karate · Python · Unity · C# · Blender

Hack4Impact *Software Engineer*

Jan 2018 – Dec 2020 (2 Years)

- Participated weekly in student-led org building robust software for nonprofits to advance social good
- Mentored 5-8 new members each semester in React frontend development and version control in GitHub
- Built a centralized virtual resource hub that equalizes and promotes information access to immigrants in Champaign (🔗)
- Developed a mobile app that gives students location-based notifications on crime, transportation, and more (🔗)

React · Flask · MongoDB

PROJECTS

Pug It Up! 🔗

- Solo developing a mobile game by self-teaching replayable level design, 2D art and animation
- Employed procedural generation for level uniqueness and difficulty scaling
- Devised a depth-first-search algorithm to find the most optimal path to victory in ~20ms

Unity · C# · Krita

HugMatch Earth 🔗

- Lead developer of *Elefano*, designed for kids to learn about global warming through multi-sensory interaction
- Collaborating with UI/UX designers to make mock-ups come to life in the game engine
- Host demos for user and social research across age range and ability, recently at Stony Brook’s Maker Faire

Unity · C# · Figma

EDUCATION

**The University of Illinois at
Urbana-Champaign**

B.S. Computer Science + Linguistics

2017 – 2020